

Machine Learning A (2026)

Home Assignment 2

Ludovico Maria Spitaleri, DMH249

Contents

1	Preprocessing (35 points)	2
1.1	Importance of Preprocessing (8 points)	2
1.2	Input Centering (9 points)	2
1.2.1	(a)	2
1.2.2	(b)	3
1.3	Input Whitening (18 points)	3
1.3.1	(a)	3
1.3.2	(b)	3
1.3.3	(c)	3
1.3.4	(d)	4
2	Hoeffding's Bound (15 points)	4
3	Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities (25 points)	5

1 Preprocessing (35 points)

1.1 Importance of Preprocessing (8 points)

The nearest neighbor to C is B, so BoL should not give him credit according to the algorithm.

$$\begin{aligned} A : \sqrt{(21 - 47)^2 + (36 - 35)^2} &\approx 26.02 \\ B : \sqrt{(21 - 22)^2 + (36 - 40)^2} &\approx 4.12 \end{aligned}$$

If instead the income is measured in dollars, A would be the nearest neighbor, so BoL should give C credit.

$$\begin{aligned} A : \sqrt{(21 - 47)^2 + (36000 - 35000)^2} &\approx 1000.34 \\ B : \sqrt{(21 - 22)^2 + (36000 - 40000)^2} &\approx 4000.00 \end{aligned}$$

This is because Euclidean distance is not scale-invariant.

1.2 Input Centering (9 points)

1.2.1 (a)

$$\begin{aligned} z_n = x_n - \bar{x} &\implies Z = X - \mathbf{1}\bar{x}^\top \\ &= X - \mathbf{1} \left(\frac{1}{N} X^\top \mathbf{1} \right)^\top \\ &= X - \frac{1}{N} \mathbf{1}\mathbf{1}^\top X \\ &= \left(I - \frac{1}{N} \mathbf{1}\mathbf{1}^\top \right) X \\ &= \gamma X \end{aligned}$$

1.2.2 (b)

1.3 Input Whitening (18 points)

1.3.1 (a)

$$\begin{aligned} \text{Var}(x_1) &= \text{Var}(\hat{x}_1) = 1 \\ \text{Var}(x_2) &= \text{Var}(\sqrt{1-\epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2) \\ &= (1-\epsilon^2)\text{Var}(\hat{x}_1) + \epsilon^2\text{Var}(\hat{x}_2) \\ &= (1-\epsilon^2)(1) + \epsilon^2(1) \\ &= 1 \\ \text{Cov}(x_1, x_2) &= \text{Cov}(\hat{x}_1, \sqrt{1-\epsilon^2}\hat{x}_1 + \epsilon\hat{x}_2) \\ &= \sqrt{1-\epsilon^2}\text{Cov}(\hat{x}_1, \hat{x}_1) + \epsilon\text{Cov}(\hat{x}_1, \hat{x}_2) \\ &= \sqrt{1-\epsilon^2}(1) + 0 \\ &= \sqrt{1-\epsilon^2} \end{aligned}$$

1.3.2 (b)

$$\begin{aligned} \hat{x}_1 = x_1, \hat{x}_2 = \frac{x_2 - \sqrt{1-\epsilon^2}x_1}{\epsilon} &\implies f(x) = \hat{w}_1(x_1) + \hat{w}_2\left(\frac{x_2 - \sqrt{1-\epsilon^2}x_1}{\epsilon}\right) \\ &= \left(\hat{w}_1 - \hat{w}_2\frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)x_1 + \left(\frac{\hat{w}_2}{\epsilon}\right)x_2 \\ \implies w_1 &= \hat{w}_1 - \hat{w}_2\frac{\sqrt{1-\epsilon^2}}{\epsilon}, \quad w_2 = \frac{\hat{w}_2}{\epsilon} \end{aligned}$$

1.3.3 (c)

$$f(\hat{x}) = \hat{x}_1 + \hat{x}_2 \implies \hat{w}_1 = 1, \hat{w}_2 = 1$$

Plugging these into our equations from section 1.3.2:

$$w_1 = 1 - \frac{\sqrt{1-\epsilon^2}}{\epsilon}, \quad w_2 = \frac{1}{\epsilon}$$

The minimum C needed for the constraint $w_1^2 + w_2^2 \leq C$ is exactly $w_1^2 + w_2^2$:

$$\begin{aligned}
C_{min} &= \left(1 - \frac{\sqrt{1-\epsilon^2}}{\epsilon}\right)^2 + \left(\frac{1}{\epsilon}\right)^2 \\
&= 1 - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{1-\epsilon^2}{\epsilon^2} + \frac{1}{\epsilon^2} \\
&= 1 - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{2-\epsilon^2}{\epsilon^2} \\
&= 1 - 2\frac{\sqrt{1-\epsilon^2}}{\epsilon} + \frac{2}{\epsilon^2} - 1 \\
&= \frac{2}{\epsilon^2} - \frac{2\sqrt{1-\epsilon^2}}{\epsilon}
\end{aligned}$$

1.3.4 (d)

As $\epsilon \rightarrow 0$, the dominant term $\frac{2}{\epsilon^2}$ prevails and $C_{min} \rightarrow \infty$. This means that as inputs become highly correlated, the weights required to construct the target function become infinitely large.

2 Hoeffding's Bound (15 points)

$$\begin{aligned}
\mathbb{E}[X] &= \frac{-2 + 0.8 + 1}{3} = \frac{-0.2}{3} = -\frac{1}{15} \\
\mathbb{E}[S] &= N\mathbb{E}[X] = -\frac{100}{15} = -\frac{20}{3} \\
X &\in [a, b] \text{ with } a = -2, b = 1 \\
\mathbb{P}(S > 22) &\implies \mathbb{P}(S - \mathbb{E}[S] > 22 - \mathbb{E}[S]) \\
&= \mathbb{P}\left(S - \mathbb{E}[S] > \frac{86}{3}\right)
\end{aligned}$$

Using Hoeffding's Inequality for sums:

$$\mathbb{P}\left(S - \mathbb{E}[S] \geq \frac{86}{3}\right) \leq \exp\left(-\frac{2 \cdot \left(\frac{86}{3}\right)^2}{100 \cdot 3^2}\right) \approx 0.1610$$

3 Illustration of Markov's, Chebyshev's, and Hoeffding's Inequalities (25 points)

3.a

1.

```
alphas = np.arange(0.5, 1 + 0.05, 0.05) # add step to endpoint to be inclusive
empirical_freq = np.array([np.mean(sample_means >= a) for a in alphas])
```

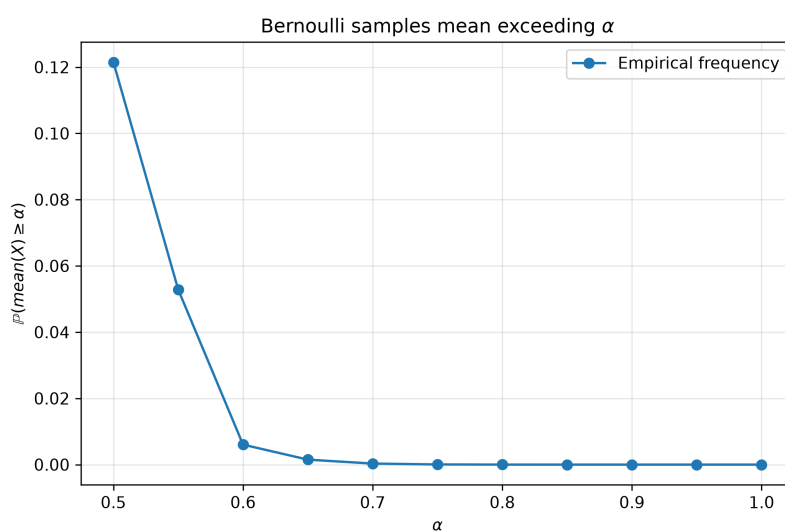


Figure 1: Empirical frequency of Bernoulli sample means exceeding α

2. Since each $X_i \in \{0, 1\}$, the sum $\sum_{i=1}^{20} X_i$ can only result in $0, 1, \dots, 20$, and the sample mean in $0, 0.05, 0.10, \dots, 1$. Therefore, taking $\alpha = 0.51$ instead of $\alpha = 0.5$ gives exactly the same probability as $\alpha = 0.55$.

3.

```
def markov_bound(alphas, expected_value):  
    return np.minimum(expected_value / alphas, 1.0)
```

```
mean_X = p
```

```
markov = markov_bound(alphas, mean_X)
```

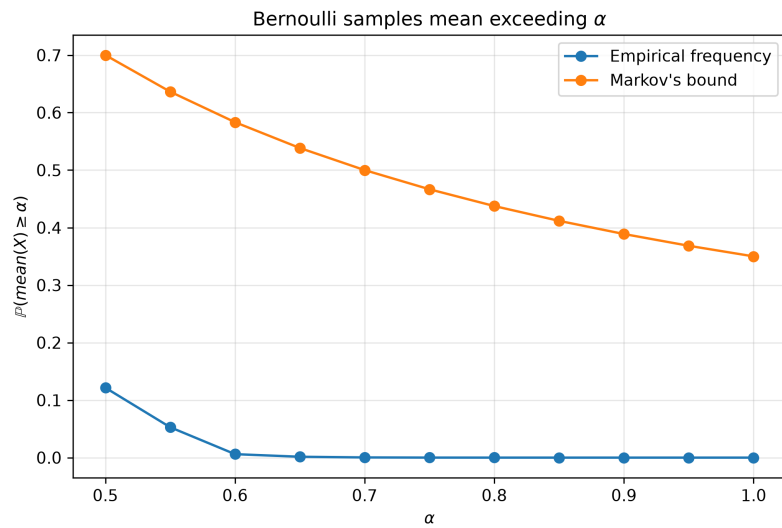


Figure 2: Empirical frequency and Markov's bound of Bernoulli sample means exceeding α

4.

```
def chebyshev_bound(alphas, mean, var, n):
    #  $P(|X - E[X]| \geq \text{eps}) \Rightarrow$  we use  $\text{eps} = \alpha - E[X]$ 
    eps = alphas - mean
    bound = np.where(eps > 0, var / (n * np.square(eps)), 1.0)
    return np.minimum(bound, 1.0)

var_X = p * (1 - p)
chebyshev = chebyshev_bound(alphas, mean_X, var_X, n)
```

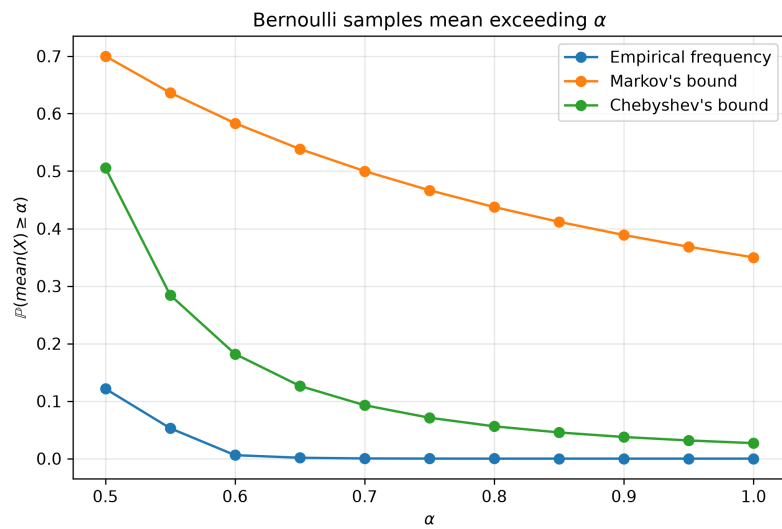


Figure 3: Empirical frequency, Markov's bound and Chebyshev's bound of Bernoulli sample means exceeding α

5.

```
def hoeffding_bound(alphas, mean, n):
    eps = alphas - mean
    bound = np.where(eps > 0, np.exp(-2 * n * np.square(eps)), 1.0)
    return np.minimum(bound, 1.0)

hoeffding = hoeffding_bound(alphas, mean_X, n)
```

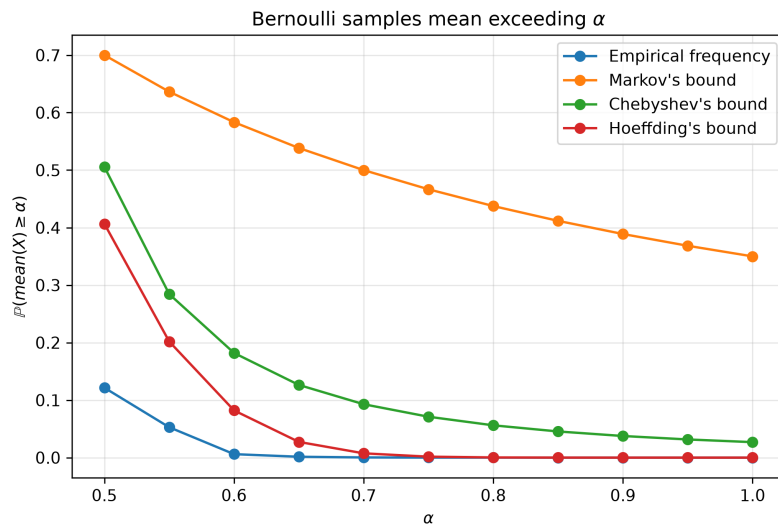


Figure 4: Empirical frequency, Markov's bound, Chebyshev's bound and Hoeffding's bound of Bernoulli sample means exceeding α

6. The empirical frequency is by far the smallest curve, decaying sharply as α grows past the mean $p = 0.35$. All three bounds lie above (or match) the true probability for every α , with some differences:

- **Markov's bound** is the loosest by a wide margin, as it only decays as $1/\alpha$.
- **Chebyshev's bound** is tighter than Markov's for most of the range, since it additionally exploits the variance of the samples, but it still decays only polynomially in α .
- **Hoeffding's bound** is by far the tightest of the three, because it decays exponentially in $n(\alpha - p)^2$.

7.

$$\sum_{i=1}^{20} X_i \sim \text{Binomial}(20, 0.35) \implies \mathbb{P}(\hat{\mu} \geq \alpha) = \mathbb{P}\left(\sum_i X_i \geq 20\alpha\right)$$

$$\begin{aligned}\mathbb{P}(\hat{\mu} \geq 1) &= \mathbb{P}(\sum_i X_i \geq 20) \approx 7.610 \times 10^{-10} \\ \mathbb{P}(\hat{\mu} \geq 0.95) &= \mathbb{P}(\sum_i X_i \geq 19) \approx 2.903 \times 10^{-8}\end{aligned}$$

3.b [optional, 0 points]